



DIGITAL FORENSICS

Practical Notebook for SOC & DFIR Analysts



Collect
Evidence



Preserve
Integrity



Analyze
Artifacts



Report
Findings



Acquire



Preserve



Examine



Analyze



Report



Present

“ In digital forensics, every bit has a story.
Our job is to preserve it, uncover it, and present it truthfully. ”

1. INTRODUCTION TO DIGITAL FORENSICS



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ WHAT IS DIGITAL FORENSICS?

Digital Forensics is the process of identifying, preserving, collecting, examining and analyzing digital evidence from electronic devices in a way that is admissible in a court of law. It helps in uncovering incidents, attacks, policy violations and cyber crimes.

★ WHY DIGITAL FORENSICS IS IMPORTANT?

- Helps in identifying the root cause of incidents.
- Preserves digital evidence in a forensically sound manner.
- Supports legal actions and compliance.
- Assists in recovering data and minimizing damage.
- Crucial for SOC, DFIR, Incident Response and Law Enforcement.

★ GOALS OF DIGITAL FORENSICS

- ✓ Preserve the integrity of evidence
- ✓ Recover as much relevant data as possible
- ✓ Analyze data to find facts and indicators
- ✓ Document the process and findings
- ✓ Present findings in a court of law

★ TYPES OF DIGITAL EVIDENCE

- Volatile Data - RAM, Running processes, Network connections, Logged-in users
- Non-Volatile Data - Files, Logs, Databases, System artifacts
- Physical Evidence - Hard drives, SSDs, USBs, Memory cards, Mobile phones
- Network Evidence - Packet captures, Firewall logs, IDS/IPS logs, DNS logs
- Cloud Evidence - Cloud storage, Logs, Snapshots, Virtual machines



SOC TIP

Always follow a forensically sound process. A small mistake in collection or handling can make the evidence useless in court!

★ TYPES OF DIGITAL FORENSICS



1. COMPUTER FORENSICS

Investigation of computers, laptops, servers and storage media.



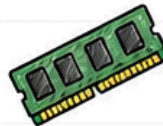
2. MOBILE FORENSICS

Extraction and analysis of data from mobile phones and tablets.



3. NETWORK FORENSICS

Capturing and analyzing network traffic and related logs.



4. MEMORY FORENSICS

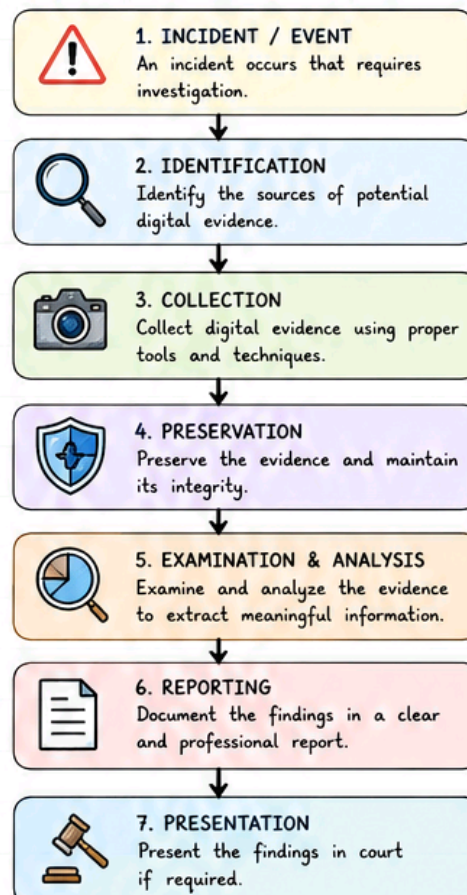
Analysis of volatile memory (RAM) to find running activities.



5. CLOUD FORENSICS

Collection and analysis of data from cloud environments and services.

★ DIGITAL FORENSIC PROCESS (OVERVIEW)



Maintain Chain of Custody through every step.

2. DIGITAL FORENSICS PROCESS & JUMP KIT



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ DIGITAL FORENSICS PROCESS

A systematic approach followed to ensure that digital evidence is collected, preserved and analyzed in a forensically sound manner.

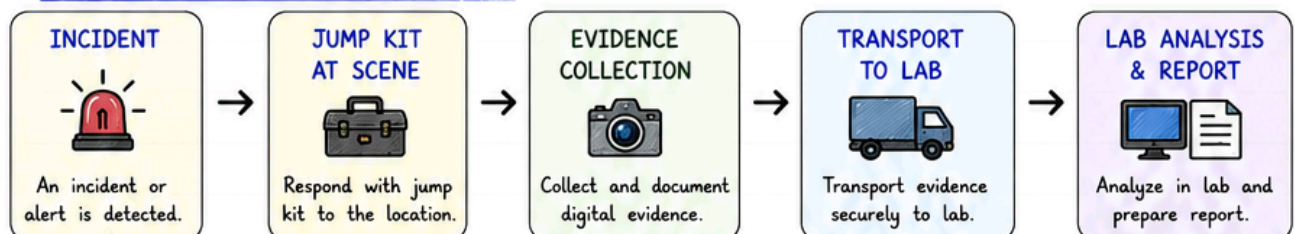
- ①  **IDENTIFICATION**
Identify potential sources of digital evidence.
- ↓
- ②  **PRESERVATION**
Preserve the evidence and maintain its integrity.
- ↓
- ③  **COLLECTION**
Collect the evidence using proper tools and procedures.
- ↓
- ④  **EXAMINATION**
Examine the evidence to extract relevant data and artifacts.
- ↓
- ⑤  **ANALYSIS**
Analyze the data to find patterns, indicators and correlations.
- ↓
- ⑥  **REPORTING**
Document the findings in a clear, accurate and professional report.
- ↓
- ⑦  **PRESENTATION**
Present the report and evidence in court if required.

★ JUMP KIT - ESSENTIAL ITEMS

A Jump Kit is a portable kit used by investigators to collect and preserve digital evidence at the incident scene.



★ INVESTIGATION WORKFLOW (HIGH LEVEL)



SOC TIP

- Always ensure the system is isolated from network before collection (if safe).
- Never analyze original evidence. Always work on forensic image / copy.
- Document everything - photos, actions, timestamps.
- Maintain integrity and chain of custody at every step.

★ KEY PRINCIPLES

- ✓ Do No Harm - Do not alter or damage evidence.
- ✓ Maintain Integrity - Ensure evidence is authentic and untampered.
- ✓ Reproducibility - Process should be repeatable.
- ✓ Legal Admissibility - Follow laws and standards (Indian IT Act, 2000).

3. EVIDENCE INTEGRITY, CHAIN OF CUSTODY & WRITE BLOCKER



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ EVIDENCE INTEGRITY

Evidence Integrity means maintaining the original state of digital evidence from the time of collection to its presentation in court. Any alteration may make the evidence inadmissible.

KEY POINTS

- ✓ Evidence must be collected properly.
- ✓ Use write blocker and validation tools.
- ✓ Create forensic images, never work on original media.
- ✓ Verify with hash values.
- ✓ Maintain Chain of Custody.

★ HASH VALUES & VERIFICATION

Hash values are unique digital fingerprints generated using algorithms.

ALGORITHM	DESCRIPTION	EXAMPLE HASH
MD5	128-bit hash	d41d8cd98f00b204e9800998ecf8427e
SHA-1	160-bit hash	da39a3ee5e6b4b0d3255bfef95601890afd80709
SHA-256	256-bit hash	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
SHA-512	512-bit hash	cf83e1357eefb8bd...3787f3053fe593da3e

HASH VERIFICATION EXAMPLE

Original Image (Disk001.E01)

SHA-256: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

After Copy (Disk001_Copy.E01)

SHA-256: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

✓ Hash Matched - Integrity Verified

★ CHAIN OF CUSTODY

Chain of Custody is a documented process that tracks the evidence from collection to final disposal.

CHAIN OF CUSTODY FORM (SAMPLE)

Case ID: _____		Evidence ID: _____			
STEP NO.	FROM (Name / Role)	TO (Name / Role)	DATE & TIME	PURPOSE	SIGNATURE
1					
2					
3					
4					
5					

IMPORTANT FIELDS EXPLAINED

- Case ID : Unique case identifier
- Evidence ID : Unique evidence identifier
- From / To : Person who transferred/received the evidence
- Date & Time : Exact time of transfer
- Purpose : Why evidence was transferred
- Signature : Proof of handover / receipt

★ WRITE BLOCKER

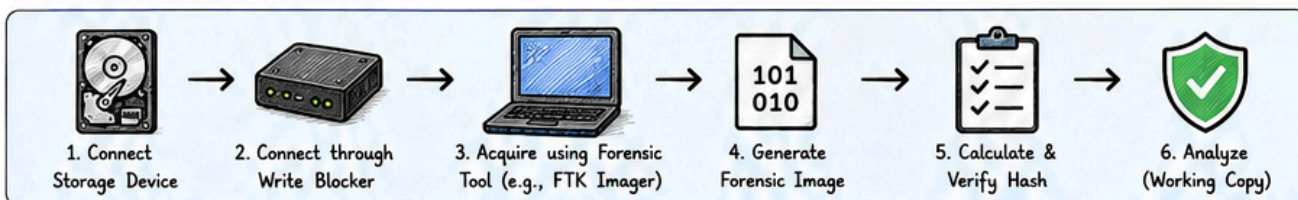
A Write Blocker is a device that prevents any data from being written to the original storage media.



BENEFITS OF WRITE BLOCKER

- ✓ Prevents modification of original data.
- ✓ Ensures evidence integrity.
- ✓ Helps in creating accurate forensic images.
- ✓ Recommended by industry standards.

★ WRITE BLOCKER WORKFLOW



★ BEST PRACTICES

- Always use write blocker while acquiring data.
- Create bit-by-bit forensic image.
- Verify hash before and after acquisition.
- Store original evidence securely.
- Document everything at every step.
- Follow chain of custody without any gaps.



SOC TIP

- Never work on original evidence.
- Always work on forensic image / copy.
- Validate tools and follow standard procedures.
- Even a small change can break the case in court!
- Document, Document, Document.

4. EVIDENCE ACQUISITION & FTK IMAGER

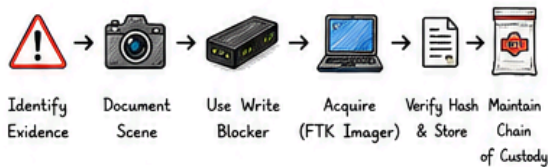


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★ TYPES OF EVIDENCE ACQUISITION

TYPE	DESCRIPTION	EXAMPLES
 Physical Acquisition	Bit-by-bit copy of entire storage media.	Hard drive, SSD, USB drive, Memory card
 Logical Acquisition	Collects only accessible data and file system information.	System files, User documents, Email, Logs
 Live Acquisition	Collect data from a running system.	RAM dump, Running processes, Network connections
 Remote / Cloud Acquisition	Collect data from remote or cloud environments.	Cloud storage, VM snapshots, Remote servers

★ EVIDENCE ACQUISITION WORKFLOW



★ FTK IMAGER OVERVIEW

FTK Imager is a free forensic tool by AccessData used to create forensic images of storage devices. It supports various formats and provides hash verification.

KEY FEATURES

- ✓ Create disk images (E01, RAW, AFF, DD).
- ✓ Verify MD5, SHA-1, SHA-256 hashes.
- ✓ Preview files and folders in the image.
- ✓ Capture memory (RAM).
- ✓ Support for compressed images.
- ✓ Generate hash set for verification.

★ DOWNLOADING & INSTALLING FTK IMAGER

- Go to: <https://accessdata.com/products-services/ftk-imager>
- Click on "Download FTK Imager".
- Choose the version for your OS (Windows).
- Install and launch FTK Imager (No license required).



SOC TIP

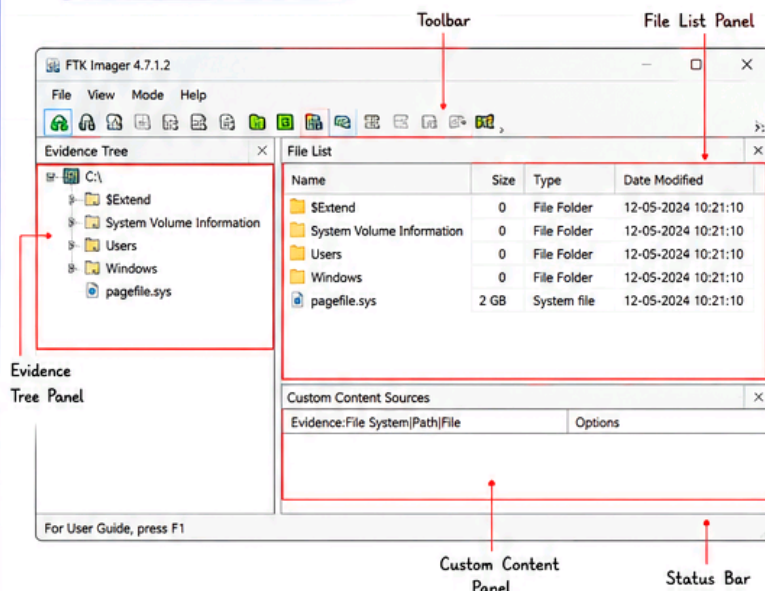
Always use FTK Imager with a write blocker. Never mount the original disk directly on your system. Work only on the forensic image.



BEST PRACTICES

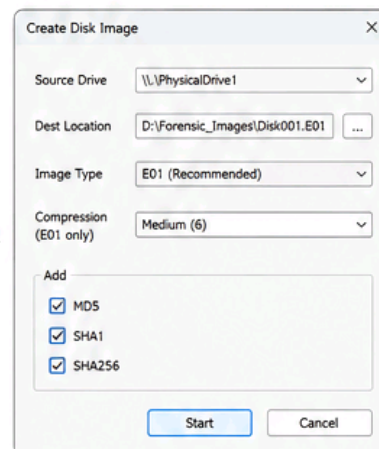
- Always document before, during and after acquisition.
- Use write blocker to prevent any accidental changes.

★ FTK IMAGER INTERFACE



★ CREATING A FORENSIC IMAGE USING FTK IMAGER

- Launch FTK Imager.
- Click on File → Create Disk Image...
- Select the source (physical drive or logical drive).
- Choose the destination location to save the image.
- Select the image type (E01 is recommended).
- Enable compression (optional).
- Click Add... to calculate hash values (MD5, SHA-1, SHA-256).
- Click Start to begin acquisition.
- After completion, verify the hashes.



★ SUPPORTED IMAGE FORMATS

FORMAT	DESCRIPTION
E01	Expert Witness Format (compressed, recommended)
RAW	Bit-by-bit copy (no compression)
AFF	Advanced Forensic Format (supports metadata)
DD	Linux / Unix raw image format

★ HASH EXAMPLE

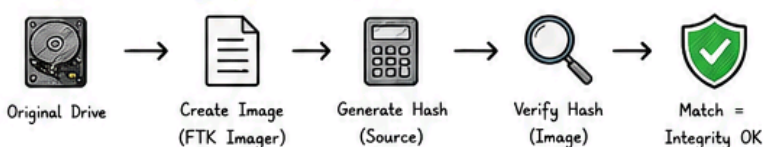
Algorithm : SHA-256

Hash Value :

e3b0c44298fc1c149afb4c8996fb924
27ae41e4649b934ca495991b7852b855

✓ Hash Matched
(Integrity Verified)

★ VERIFICATION AFTER IMAGING



5. DISK IMAGING & DELETED FILE RECOVERY

★ DISK IMAGING

Disk imaging is the process of creating an exact bit-by-bit copy (forensic image) of a storage device without altering the original data.

Why Disk Imaging?

- ✓ Preserves original evidence.
- ✓ Allows multiple analysis without affecting data.
- ✓ Accepted in court as a reliable method.
- ✓ Protects data from accidental modification.

COMMON IMAGE FORMATS

FORMAT	DESCRIPTION
E01	Expert Witness Format (compressed, recommended)
RAW	Bit-by-bit copy (no compression)
AFF	Advanced Forensic Format (supports metadata)
DD	Linux / Unix raw image format

★ TYPES OF FORENSIC IMAGES

- Full Disk Image - Entire drive is copied.
- Partition Image - Only selected partition is copied.
- Logical Image - Files and folders only.
- Custom Image - Specific sectors / data.

★ DELETED FILE RECOVERY SOURCES

- Recycle Bin
- Unallocated Space
- File Slack Space
- Shadow Copies
- Temp Files
- Page File / Swap File



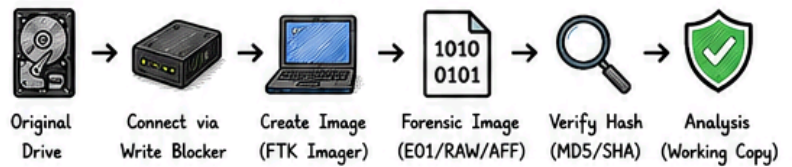
SOC TIP

Always work on a forensic image (copy), never on the original drive. Even a small change in timestamps can break the case in court!

★ DELETED FILE RECOVERY TECHNIQUES

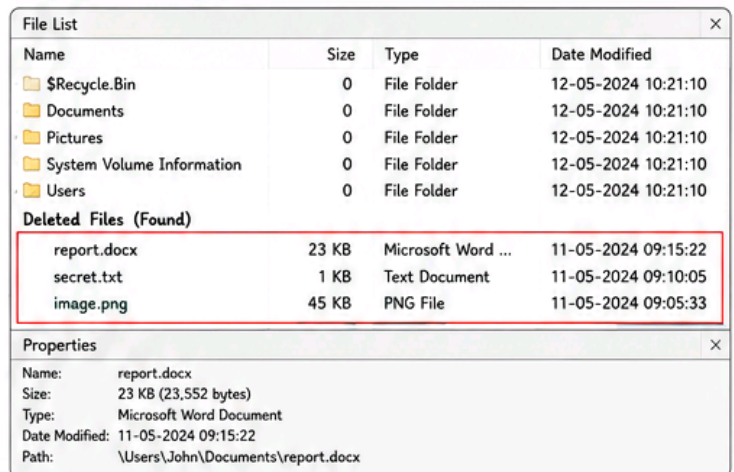
TECHNIQUE	DESCRIPTION
Recycle Bin	Files deleted to recycle bin can be restored.
Unallocated Space Recovery	Files not overwritten can be recovered.
Slack Space Analysis	Data remnants in slack space of clusters.
File Carving	Search for file signatures in raw data.
System Restore / Shadow Copy	Recover previous versions of files.

★ DISK IMAGING WORKFLOW



★ DELETED FILE RECOVERY EXAMPLE (FTK IMAGER)

- 1 Open the forensic image in FTK Imager.
- 2 Go to File → Add Evidence Item → Image File.
- 3 Expand the image in the left pane.
- 4 Right-click on the folder / drive.
- 5 Click "Export Files" or "Save Files" to recover deleted files.



Recovered files are usually marked in red or placed in "Deleted Files" folder in FTK Imager.

★ FILE CARVING (SIGNATURE BASED RECOVERY)

File carving searches for known file signatures in raw data to recover files that have no file system entries.

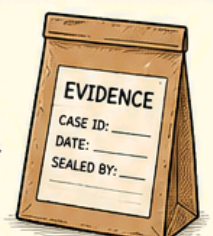
Common File Signatures:

FILE TYPE	HEADER (HEX)	FOOTER (HEX)	DESCRIPTION
JPEG	FF D8 FF	FF D9	Image file
PNG	89 50 4E 47	49 45 4E 44 AE 42 60 82	Image file
PDF	25 50 44 46	25 25 45 4F 46	Document file
ZIP	50 4B 03 04	50 4B 05 06	Compressed file



BEST PRACTICES

- ✓ Always use a write blocker before imaging.
- ✓ Verify hash before and after imaging.
- ✓ Store original media in a safe evidence bag.
- ✓ Recover files only from forensic image (never from original).
- ✓ Document every action during imaging and recovery.
- ✓ Maintain chain of custody at every step.



6. WINDOWS FORENSICS & ARTIFACTS



BLUETEAMERS
DEFEND. DETECT. RESPOND.

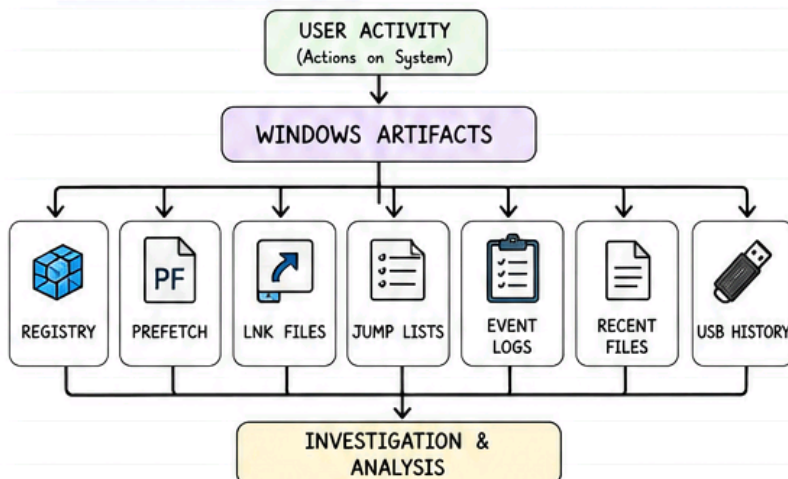
★ WHAT ARE WINDOWS ARTIFACTS?

Artifacts are the digital footprints left behind by user and system activities on a Windows system. These artifacts help investigators reconstruct what happened, when it happened and how it happened.

WHY ARE ARTIFACTS IMPORTANT?

- ✓ Help in understanding user actions.
- ✓ Useful in timeline reconstruction.
- ✓ Reveal attacker persistence & movement.
- ✓ Support incident response and threat hunting.
- ✓ Provide valuable evidence in court.

★ WINDOWS ARTIFACTS FLOW



★ IMPORTANT WINDOWS ARTIFACTS – REAL EXAMPLES

EVENT VIEWER (SECURITY LOG)

Event Viewer (Local)	Security	Number of events: 25,144
> Custom Views	Keywords	Date and Time
> Windows Logs		Event ID
Application	Audit Success	12-05-2024 10:21:10 4624
Security	Audit Failure	12-05-2024 10:20:55 4625
Setup	Audit Success	12-05-2024 10:20:41 4688
System	Audit Success	12-05-2024 10:20:35 4672
Forwarded Events	Audit Success	12-05-2024 10:20:10 4720
Applications and Services Logs		

REGISTRY EDITOR

Computer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run		
	Name	Data
> Computer		
> HKEY_CLASSES_ROOT	(Default)	(value not set)
> HKEY_CURRENT_USER	OneDrive	"C:\Users\Public\...\OneDrive.exe"
> Software		
> Microsoft		
> Windows		
> CurrentVersion		
> Run		

PREFETCH FOLDER

Name	Date Modified
CHROME.EXE-3A7F2B4C.pf	12-05-2024 10:15
WINWORD.EXE-1A2B3C4D.pf	12-05-2024 10:14
EXPLORER.EXE-5D6E7F8A.pf	12-05-2024 10:12
NOTEPAD.EXE-9B8C7D6E.pf	12-05-2024 10:10

★ MAJOR WINDOWS ARTIFACTS – QUICK REFERENCE

ARTIFACT	PURPOSE	DEFAULT LOCATION	INVESTIGATION VALUE
Registry	Stores system configuration, user settings, installed programs, run keys.	C:\Windows\System32\config\ C:\Users\<User>\NTUSER.DAT	Find persistence, installed apps, recent activity, user behavior.
Prefetch	Stores info about executed programs for performance.	C:\Windows\Prefetch\	Identify executed programs and timeline.
LNK Files	Shortcut files that store path, execution time and more.	C:\Users\<User>\AppData\Roaming\Microsoft\Windows\Recent\	Reveal recently opened files and user activity.
Jump Lists	Tracks recently opened files, folders and programs.	C:\Users\<User>\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\	Useful for timeline reconstruction and user behavior.
Event Logs	Logs system, security, application and setup events.	Event Viewer (eventvwr.msc) C:\Windows\System32\winevt\Logs\	Track logons, logoffs, errors, process creation and more.
Recent Files	List of files recently opened by the user.	C:\Users\<User>\AppData\Roaming\Microsoft\Windows\Recent\	Identify accessed documents and timeline.
USB History	Records of USB devices connected to the system.	Registry -> SYSTEM\CurrentControlSet\Enum\USBSTOR\	Find connected USB devices and usage timeline.
Recycle Bin	Stores deleted files before permanent removal.	C:\\$Recycle.Bin\	Recover deleted files and artifacts.



SOC TIP

- Always collect volatile data (RAM) before analyzing artifacts.
- Correlate multiple artifacts to build a strong timeline.
- Artifacts can be modified or deleted by attackers.
- Use timeline tools like KAPE, Plaso (log2timeline), or EZTools.
- Document every artifact and its source.



USEFUL TOOLS FOR ARTIFACT ANALYSIS

- ✓ Registry Explorer - Analyze registry hives.
- ✓ Prefetch Viewer (WinPrefetchView) - Analyze Prefetch files.
- ✓ LECmd / KAPE - Collect artifacts.
- ✓ EZTools - All-in-one Windows forensics tool.
- ✓ Event Viewer - Analyze Windows logs.
- ✓ Timeline Explorer / Plaso - Create timeline from artifacts.



★ REMEMBER

Artifacts are the silent witnesses of every action on a system. The more artifacts you collect and analyze, the closer you get to the truth.



7. RAM FORENSICS & MEMORY ANALYSIS



BLUETEAMERS

DEFEND. DETECT. RESPOND.

★ WHY RAM FORENSICS?

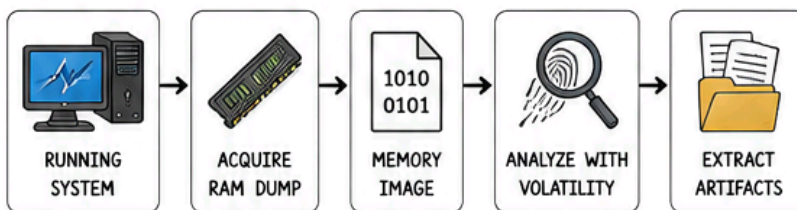
RAM (Random Access Memory) contains volatile data about running processes, network connections, loaded modules, logged-on users, malware artifacts and more. It can disappear once the system is powered off.



KEY POINTS

- ✓ Memory is volatile – capture first.
- ✓ RAM provides real-time view of the system.
- ✓ Helps in malware analysis and incident investigation.
- ✓ Essential for detecting fileless attacks.
- ✓ Capture RAM before shutting down.

★ MEMORY FORENSICS WORKFLOW



WHAT WE CAN EXTRACT FROM MEMORY?

- Running Processes
- Network Connections
- Loaded DLLs & Drivers
- Logged-on Users
- Command History
- Malware Artifacts
- Registry Hives (in memory)
- Active Sessions

★ MEMORY ACQUISITION TOOLS

TOOL	TYPE	DESCRIPTION
DumpIt	Free	Command line tool to create RAW memory dump.
WinPMEM	Free	High performance memory acquisition tool.
FTK Imager	Free	Can capture RAM using "Add Evidence Item".
Belkasoft RAM Capturer	Free	Easy GUI based RAM capture tool.
Magnet RAM Capture	Free	User-friendly tool for RAM acquisition.

★ RAM ACQUISITION METHODS

- ✓ Use a trusted tool.
- ✓ Run as Administrator.
- ✓ Save the memory image in a safe location.
- ✓ Verify hash of the memory image.
- ✓ Analyze with Volatility.



SOC TIP

Always capture memory before shutting down the system. Once power is off, volatile data is lost forever!

★ RAM ACQUISITION EXAMPLE (DUMPTIT)

1. DumpIt.exe
2. Select a destination path
3. Click "Dump"
4. Memory image (raw) will be created



★ VOLATILITY FRAMEWORK

Volatility is the most popular open-source memory forensics framework used to analyze memory dumps and extract artifacts.

```
vol.py -f memory.raw imageinfo
```

```
-----
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : Win10x64_19041
AS Layer1 : IA32PagedMemory (Kernel AS)
AS Layer2 : FileAddressSpace (C:\memory.raw)
PAE type : No PAE
DTB : 0x1a0000L
KDBG : 0xfffff801'5c3d1b0L
Number of Processors : 8
Image Type (Service Pack) : 0
KPCR for CPU 0 : 0xfffff801'5c56000L
KUSER_SHARED_DATA : 0xfffff700'0000000L
Image date and time : 2024-05-12 10:21:33 UTC+0000
Image local date and time : 2024-05-12 15:51:33 +0530
```

★ SAMPLE OUTPUT (pslist)

```
vol.py -f memory.raw pslist
```

Offset(V)	Name	PID	PPID	Thds	Hnds	Sess	Wow64	Start
0xfffffa8002a0a1b0	System	4	0	119	570	N/A	False	2024-05-12
0xfffffa8002ac4400	smss.exe	312	4	2	29	N/A	False	2024-05-11
0xfffffa8002ad5000	csrss.exe	440	432	13	349	0	False	2024-05-11
0xfffffa8002ad6000	wirinit.exe	506	432	4	77	0	False	2024-05-12
0xfffffa8002ae3000	services.exe	588	506	9	210	0	False	2024-05-12
0xfffffa8002b2d000	svchost.exe	732	588	27	612	0	False	2024-05-12
...								



BEST PRACTICES

- ✓ Acquire RAM before any shutdown or reboot.
- ✓ Use trusted tools to avoid corruption.
- ✓ Store memory image in a secure location.
- ✓ Verify hash of memory image.
- ✓ Document every step of the acquisition.
- ✓ Use the correct profile in Volatility.

★ IMPORTANT VOLATILITY COMMANDS

COMMAND	DESCRIPTION	USAGE EXAMPLE
imageinfo	Shows information about the memory image.	vol.py -f memory.raw imageinfo
pslist	Lists all running processes.	vol.py -f memory.raw pslist
pstree	Shows process tree.	vol.py -f memory.raw pstree
netscan	Shows network connections.	vol.py -f memory.raw netscan
dlllist	Lists loaded DLLs for processes.	vol.py -f memory.raw dlllist
cmdline	Shows command line arguments.	vol.py -f memory.raw cmdline
handles	Shows open handles.	vol.py -f memory.raw handles
connections	Shows network connections (with PID).	vol.py -f memory.raw connections
malfind	Detects injected code in memory.	vol.py -f memory.raw malfind
filescan	Scans for file objects in memory.	vol.py -f memory.raw filescan



TIP FOR SOC ANALYSTS

Memory analysis helps detect advanced threats, fileless malware, in-memory injections and attacker activities not visible in disk artifacts.

8. WINDOWS EVENT LOGS & TIMELINE ANALYSIS



BLUETEAMERS

DEFEND. DETECT. RESPOND.

★ WHAT ARE WINDOWS EVENT LOGS?

Windows Event Logs are records of events that occur on a system. They contain valuable information about user logons, process execution, system changes, security events and more. Useful for incident response, threat hunting and forensics.



WHY ARE EVENT LOGS IMPORTANT?

- ✓ Track user and system activities.
- ✓ Identify suspicious or malicious actions.
- ✓ Correlate events for timeline analysis.
- ✓ Support incident response and forensics.

★ WINDOWS EVENT LOG CATEGORIES

Security Log



Records security related events like logon, logoff, privilege use, policy changes, etc.

Location:

Event Viewer →
Windows Logs →
Security

System Log



Records system activities like driver loading, system boot, shutdown, errors, etc.

Location:

Event Viewer →
Windows Logs →
System

Application Log



Records events from installed applications and services.

Location:

Event Viewer →
Windows Logs →
Application

Setup Log



Records events related to Windows setup, installation and updates.

Location:

Event Viewer →
Windows Logs →
Setup

★ IMPORTANT EVENT IDS (SECURITY LOG)

EVENT ID	DESCRIPTION	USE CASE
4624	Successful Logon	User logged in successfully
4625	Failed Logon	Failed login attempt
4634	Logoff	User logged off
4648	Logon with Explicit Credentials	Credentials were explicitly used to log on
4672	Special Privileges Assigned	Admin/privileged account logged in
4688	Process Creation	A new process was created
4720	User Account Created	A new user account was created
4726	User Account Deleted	A user account was deleted
4732	Member Added to Group	User added to security group
1102	Log Cleared	Security log was cleared

★ EVENT LOG TIMELINE INVESTIGATION FLOW



ALERT /
INCIDENT



USER
IDENTIFICATION
(4624)



PROCESS
CREATION
(4688)



PARENT PROCESS
RELATIONSHIP
(PSTREE)



TIMELINE
CORRELATION



CONCLUSION &
REPORTING

- ✓ Start with the alert or suspicious activity.
- ✓ Identify the user who performed the action (4624).
- ✓ Track the processes created by the user (4688).
- ✓ Identify parent-child process relationship.
- ✓ Correlate all related events to build a complete timeline.
- ✓ Draw conclusions and document findings.

★ REAL EXAMPLE - WINDOWS EVENT VIEWER (SECURITY LOG)

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Subscriptions

Security Number of events: 35,214

Keywords	Date and Time	Event ID	Task Category
Audit Success	05-12-2024 10:21:10	4624	Logon
Audit Success	05-12-2024 10:21:15	4688	Process Creation
Audit Success	05-12-2024 10:21:16	4672	Special Logon
Audit Failure	05-12-2024 10:21:18	4625	Logon
Audit Success	05-12-2024 10:21:22	4634	Logoff
Audit Success	05-12-2024 10:21:35	4720	User Account Management

Event 4624, Microsoft Windows security auditing.

General Details

An account was successfully logged on.

Log Name: Security
Source: Microsoft Windows security auditing.
Event ID: 4624
Level: Information
User: DESKTOP-1A2B3C\saiprasanna
Logon ID: 0x3E7
Task Category: Logon
Keywords: Audit Success
Time Created: 05-12-2024 10:21:10

★ BUILDING A TIMELINE - WHAT TO LOOK FOR?

- ✓ Successful and failed logon attempts (4624, 4625)
- ✓ Process creation and command execution (4688)
- ✓ Privilege escalation events (4672)
- ✓ New user or group changes (4720, 4732)
- ✓ Policy and configuration changes
- ✓ Log clearing attempts (1102)
- ✓ System startup and shutdown events

★ TOOLS FOR LOG ANALYSIS

- Event Viewer (Built-in)
- Timeline Explorer (Plaso)
- Chainsaw
- KAPE
- ELK / Splunk
- Sysmon (For Detailed Logging)



SOC TIP

Always correlate logs from multiple sources (Security, System, Application and Sysmon) to build an accurate timeline. Attackers may try to clear logs or tamper with timestamps.



BEST PRACTICES

- ✓ Enable auditing policies (Logon, Process, Object Access, etc.).
- ✓ Synchronize system time using NTP.
- ✓ Collect logs centrally (SIEM) for better visibility.
- ✓ Regularly review logs for suspicious activity.
- ✓ Retain logs for required period as per policy.
- ✓ Protect logs from tampering and deletion.
- ✓ Document all findings during investigations.
- ✓ Use timeline analysis to understand the full picture.

9. BROWSER FORENSICS



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ WHAT IS BROWSER FORENSICS?

Browser forensics is the process of collecting, preserving and analyzing data stored by web browsers. This data helps investigators identify user activity, track online behavior, and build a timeline of events.

WHY IS IT IMPORTANT?

- ✓ Track user web activity and visits
- ✓ Recover deleted history and downloads
- ✓ Find saved credentials and cookies
- ✓ Identify malicious or suspicious websites
- ✓ Support incident response and investigations

★ BROWSER FORENSICS WORKFLOW



TYPES OF DATA WE CAN EXTRACT

- Browsing History
- Downloads
- Cookies
- Cache
- Saved Passwords
- Form Data / Autofill
- Bookmarks / Favorites
- Session Information
- Extensions
- Search Queries
- Active Tabs (Sessions)
- Web Artifacts (SQLite DB)

★ MAJOR BROWSERS & DATA LOCATIONS

BROWSER	HISTORY LOCATION	COOKIES LOCATION	DOWNLOADS LOCATION	LOGIN DATA / PASSWORDS
Google Chrome	%LOCALAPPDATA%\Google\Chrome\User Data\Default\History	%LOCALAPPDATA%\Google\Chrome\User Data\Default\Network\Cookies	%USERPROFILE%\Downloads (History stored in History DB)	%LOCALAPPDATA%\Google\Chrome\User Data\Default>Login Data (SQLite DB)
Microsoft Edge	%LOCALAPPDATA%\Microsoft\Edge\User Data\Default\History	%LOCALAPPDATA%\Microsoft\Edge\User Data\Default\Network\Cookies	%USERPROFILE%\Downloads (History stored in History DB)	%LOCALAPPDATA%\Microsoft\Edge\User Data\Default>Login Data (SQLite DB)
Mozilla Firefox	%APPDATA%\Mozilla\Firefox\Profiles\<profile>\places.sqlite	%APPDATA%\Mozilla\Firefox\Profiles\<profile>\cookies.sqlite	%USERPROFILE%\Downloads (History stored in places.sqlite)	%APPDATA%\Mozilla\Firefox\Profiles\<profile>\logins.json

★ REAL EXAMPLE - CHROME HISTORY (HISTORY DB)

id	url	title	visit_count	last_visit_time
1	https://www.google.com/	Google	15	13489012345678
2	https://mail.google.com/	Gmail	8	13489012355678
3	https://github.com/	GitHub	5	13489012365678
4	https://www.youtube.com/	YouTube	12	13489012375678
5	https://www.blueteamers.io/	Blueteamers	3	13489012385678

Note: Timestamps are stored in microseconds since 1601 (UTC).
Convert using: $\text{ChromeTime} = (\text{WebKitTime} / 1000000) - 11644473600$

★ BROWSER ARTIFACTS OVERVIEW

ARTIFACT	DESCRIPTION	LOCATION / FILE
History	Websites visited by the user	History (SQLite DB)
Cookies	Stores website cookies & sessions	Cookies (SQLite DB)
Cache	Temporary files (images, scripts, etc.)	Cache Folder
Downloads	Files downloaded by the user	History DB + Downloads Folder
Passwords	Saved webpage credentials	Login Data / logins.json
Bookmarks	Saved bookmarks / favorites	Bookmarks (JSON / DB)
Sessions	Active tabs and session info	Sessions / Current Tabs
Search Queries	Search terms typed in browser	History / Keyword Search DB
Extensions	Installed extensions & IDs	Preferences / Local Storage
Web Artifacts	SQLite, Local Storage, IndexedDB	various DB files

★ TOOLS FOR BROWSER FORENSICS

- Browser History Examiner (BHE)**
Analyze Chrome, Edge, Firefox data
- DB Browser for SQLite**
View and extract SQLite database data
- Autopsy**
Open source DFIR tool for artifact analysis
- Belkasoft Evidence Center**
Powerful commercial DFIR platform
- Bulk Extractor / RegRipper**
Extract browser related artifacts
- Strings / Timeline Explorer (Plaso)**
Build timeline from browser artifacts

SAMPLE FINDINGS - INVESTIGATION

- ✓ User visited suspicious site:
[http://malicious-site\[.\]com](http://malicious-site[.]com)
- ✓ Downloaded file: setup.exe
from 10.25.2024 14:32:10
- ✓ Logged in to email at 10.25.2024 14:40:05
- ✓ Searched for: "how to hack wifi"
at 10.25.2024 14:41:22
- ✓ Saved password for ftp.example.com
(username: admin)

SOC TIP

- ✓ Always create a forensic image before analyzing the system.
- ✓ Browser data can be deleted by users, use unallocated space & artifacts to recover.
- ✓ Correlate browser timeline with event logs, downloads and emails.
- ✓ Use multiple tools for accurate and complete analysis.

★ BEST PRACTICES

- ✓ Always acquire disk image before starting analysis.
- ✓ Collect all browser profiles (multiple users).
- ✓ Analyze SQLite databases carefully.
- ✓ Recover deleted data using carving techniques.
- ✓ Correlate browser artifacts with system timeline.
- ✓ Document every finding with timestamps.
- ✓ Preserve evidence integrity and maintain chain of custody.
- ✓ Use hashing (MD5/SHA256) to verify evidence.

10. MULTIMEDIA FORENSICS & STEGANOGRAPHY



BLUETEAMERS

DEFEND. DETECT. RESPOND.

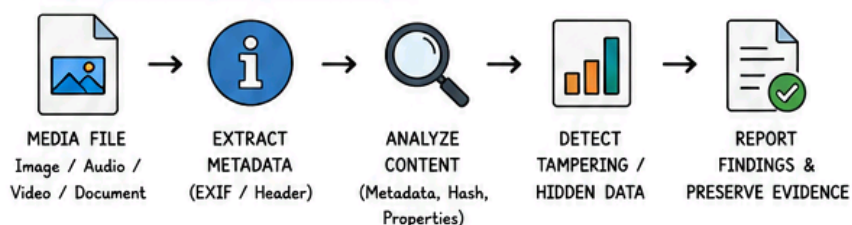
★ WHAT IS MULTIMEDIA FORENSICS?

Multimedia forensics involves extracting, analyzing and interpreting data from images, audio, video and documents. It helps investigators uncover hidden information, verify authenticity and build timelines.

WHY IS IT IMPORTANT?

- ✓ Verify authenticity of media files.
- ✓ Extract hidden metadata and GPS info.
- ✓ Detect image tampering and manipulation.
- ✓ Recover hidden data using steganalysis.
- ✓ Provide strong digital evidence in court.

★ MULTIMEDIA FORENSICS WORKFLOW



TYPES OF MULTIMEDIA FILES ANALYZED



★ METADATA & EXIF INFORMATION

Metadata is "data about data". EXIF (Exchangeable Image File Format) stores valuable info captured by cameras and devices.

COMMON EXIF FIELDS

FIELD	DESCRIPTION
Camera Make	Manufacturer of the device
Camera Model	Model of the camera / device
Date/Time	Date and time when the image was taken
GPS Coordinates	Latitude and Longitude (if GPS enabled)
Exposure Time	Duration the shutter was open
Focal Length	Lens focal length used
ISO Speed	Light sensitivity of the camera
Software	Software used to edit or modify
Resolution	Image dimensions (width x height)

★ REAL EXAMPLE - EXIF DATA (ExifTool)

```
$ exiftool image.jpg
ExifTool Version Number      : 12.76
File Name                    : image.jpg
Directory                    : C:/Evidence/Images
File Size                    : 2.45 MB
File Modification Date/Time  : 2024:05:10 14:32:10+05:30
File Access Date/Time       : 2024:05:10 14:32:12+05:30
File Inode Change Date/Time  : 2024:05:10 14:32:12+05:30
Camera Make                  : Canon
Camera Model                 : Canon EOS 750D
Date/Time Original          : 2024:05:09 10:24:15
Create Date                 : 2024:05:09 10:24:15
Exposure Time               : 1/250
F Number                    : 5.6
ISO                         : 400
Focal Length                : 50.0 mm
GPS Latitude                 : 17 deg 23' 45.23" N
GPS Longitude               : 78 deg 27' 11.65" E
Software                    : Adobe Photoshop 24.2
....
```

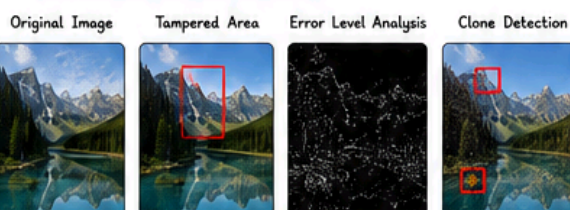
TOOLS FOR METADATA ANALYSIS

- ExifTool**
(Command line tool)
- ExifViewer**
(GUI tool)
- Metadata2Go**
(Online tool)
- Jeffrey's Image Metadata Viewer**
- Hachoir**
(Open source)

SOC TIP

Always check metadata for timestamps, GPS and device info. Attackers may remove or modify this data.

★ IMAGE TAMPERING DETECTION



TAMPERING INDICATIONS

- ✓ Inconsistent lighting and shadows
- ✓ Different compression levels
- ✓ Cloned / repeated regions
- ✓ Irregular edges or pixel patterns
- ✓ Metadata inconsistencies
- ✓ EXIF timestamp mismatch
- ✓ Histogram anomalies

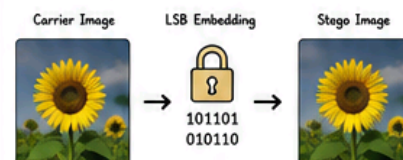
★ STEGANOGRAPHY - HIDING DATA IN MEDIA

Steganography hides secret data inside images, audio or videos without visible changes. It is used by attackers to hide malicious payloads or messages.

COMMON STEGANOGRAPHY TOOLS

- StegHide**
(Hide data in images)
- StegDetect**
(Detect hidden data)
- zsteg**
(LSB steganalysis tool)
- OutGuess**
(Embed data in images)

LSB STEGANOGRAPHY EXAMPLE



HOW IT WORKS?

Least Significant Bits (LSB) of pixels are changed to store hidden data. Human eyes can't detect the difference.

★ OTHER ANALYSIS TECHNIQUES

- Hashing**
(MD5, SHA1, SHA256 to verify integrity)
- File Signature Analysis**
(Check header and file structure)
- Audio Spectrogram Analysis**
(Detect hidden audio data)
- Video Frame Analysis**
(Extract frames and analyze anomalies)

★ COMMON HASH ALGORITHMS

ALGORITHM	OUTPUT SIZE	DESCRIPTION
MD5	128-bit	Fast but weak (collision possible)
SHA-1	160-bit	Stronger than MD5 (deprecated)
SHA-256	256-bit	Strong and widely used
SHA-512	512-bit	Very strong and secure

- ✓ Always generate hash before and after analysis to ensure data integrity.

★ BEST PRACTICES

- ✓ Always create forensic image of media.
- ✓ Work on copies, never on originals.
- ✓ Extract metadata and hash the file.
- ✓ Document every step and tool used.
- ✓ Preserve original integrity and maintain chain of custody.



11. ANTI-FORENSICS TECHNIQUES



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ WHAT IS ANTI-FORENSICS?

Anti-forensics techniques are used by attackers to evade detection and make it difficult for investigators to collect, recover and analyze digital evidence. A skilled DFIR analyst can detect, counter and recover these activities.



WHY DO ATTACKERS USE IT?

- ✓ Hide their identity and actions
- ✓ Delay detection and response
- ✓ Destroy or modify evidence
- ✓ Evade accountability

★ ANTI-FORENSICS TECHNIQUES OVERVIEW



LOG CLEANING
Clearing event logs, shell history, etc.



TIMESTAMP MANIPULATION
Modify file timestamps to mislead timeline.



ALTERNATE DATA STREAMS (ADS)
Hide malicious data inside files.



ENCRYPTION
Encrypt files or disks to prevent analysis.



SECURE DELETE
Overwriting files to prevent recovery.



METADATA REMOVAL
Remove metadata from files, images, documents.



TIMESTAMPING
Change timestamps of files to match benign files.



DISK WIPING
Wipe free space or entire disk securely.



STEGANOGRAPHY
Hide data inside images, audio, video files.



COMMAND HISTORY REMOVAL
Clear shell / terminal history.

★ ANTI-FORENSICS DETECTION & RECOVERY FLOW



HOW INVESTIGATORS COUNTER ANTI-FORENSICS?

- ✓ Collect volatile data (RAM) quickly.
- ✓ Validate findings with timelines.
- ✓ Use reliable tools to recover artifacts.
- ✓ Use memory, logs, network, registry.
- ✓ Analyze raw disk, unallocated space.
- ✓ Document every step and finding.
- ✓ Correlate multiple data sources.

★ TOOLS TO DETECT & RECOVER ARTIFACTS



Autopsy / The Sleuth Kit
File recovery & analysis



AccessData FTK
Advanced forensics suite



FTK Imager
Create forensics image, verify integrity



Strings
Extract readable strings



Registry Explorer
Analyze registry & artifacts



StegDetect / zsteg
Detect steganography in files



Bulk Extractor
Extract data from disk image



Wireshark
Network artifact analysis



Volatility
Memory analysis & artifact recovery



Recuva / PhotoRec
File recovery tools

★ COMMON ANTI-FORENSICS TECHNIQUES & DETECTION

TECHNIQUE	DESCRIPTION	DETECTION / RECOVERY
Log Cleaning	Clearing Windows Event Logs, Sysmon Logs, etc.	Check for Event ID 1102, gaps in logs, log backups
Timestamp Manipulation	Changing file timestamps (Created/Modified/Accessed)	Timeline analysis, Compare with MFT, \$SI
ADS (Alternate Data Streams)	Hiding data in NTFS alternate streams	Use tools: streams.exe, Invoke-ADS, FTK
Encryption	Encrypting files/folders or entire drives	Detect encrypted files, Check for ransom notes
Secure Delete	Overwriting files to prevent recovery	File carving, Check unallocated space
Metadata Removal	Removing EXIF, author, timestamps, GPS, etc.	ExifTool, metadata analysis to recover info
Timestamping	Modifying timestamps to match other files	Cross-check with MFT, timeline, logs
Disk Wiping	Wiping free space or entire disk	Check for wiping tools usage, patterns in MFT
Steganography	Hiding data in images, audio, video	Use StegDetect, Binwalk, strings, zsteg
History Removal	Clearing browser, shell, and app history	Recover from cache, SQLite, \$UsnJrnl, Prefetch

★ REAL EXAMPLE - TIMESTAMP MANIPULATION

```
C:\> dir /T:W
Volume in drive C has no label.
Volume Serial Number is 1A2B-3C4D

Directory of C:\Evidence

06/01/2024  08:30 AM    <DIR>          ..
06/01/2024  08:30 AM    <DIR>          .
01/01/2024  12:00 AM                4,096  legit.docx
01/01/2024  12:00 AM               15,872  report.pdf
                2 File(s)              19,968 bytes
                2 Dir(s)         120,456,789,504 bytes free
```

INVESTIGATOR FINDING:

File timestamps are suspicious (set to 01/01/2024).
MFT and \$SI timestamps show actual creation on 06/01/2024.
Conclusion: Timestamps were manipulated to hide real activity.



SOC TIP

- ✓ Always capture volatile data before shutdown.
- ✓ Do not trust timestamps - validate with MFT, logs, and artifacts.
- ✓ Analyze unallocated space for deleted or hidden data.
- ✓ Correlate artifacts from multiple sources for accuracy.
- ✓ Document every step of investigation.



★ BEST PRACTICES

- ✓ Acquire evidence using write blockers.
- ✓ Verify hashes before and after acquisition.
- ✓ Preserve original evidence and chain of custody.
- ✓ Analyze on forensic copies, never originals.
- ✓ Use multiple tools to validate findings.
- ✓ Maintain a detailed timeline of events.
- ✓ Report findings clearly with proof.



“

Attackers try to hide evidence, but digital footprints remain.
A thorough DFIR analyst turns hidden artifacts into the truth.

12. WEBSITE CRIME INVESTIGATION & OSINT



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ WHAT IS WEBSITE CRIME INVESTIGATION?

Website crime investigation is the process of identifying, analyzing and gathering intelligence about malicious websites used for cyber attacks, fraud, phishing, malware distribution, command & control and other illegal activities.

COMMON WEBSITE CRIMES



Website Defacement



Phishing Websites



Fake Login Pages



Malware Hosting

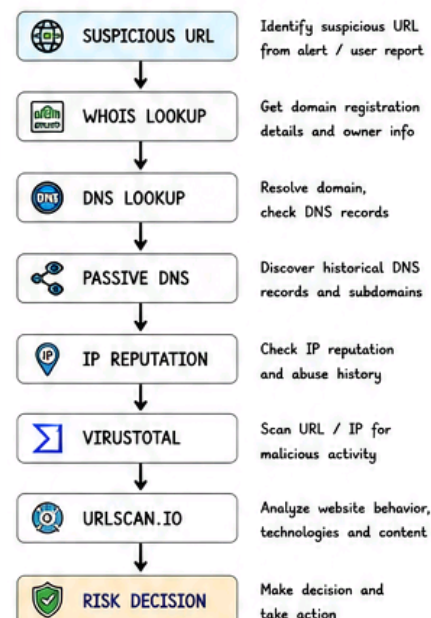


Scam / Fraud Websites



C2 Servers / Botnet Panels

★ WEBSITE INVESTIGATION WORKFLOW

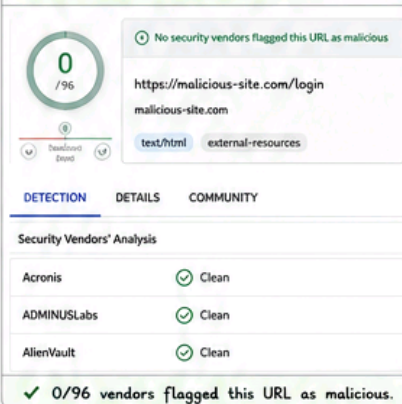


★ INVESTIGATION TOOLS

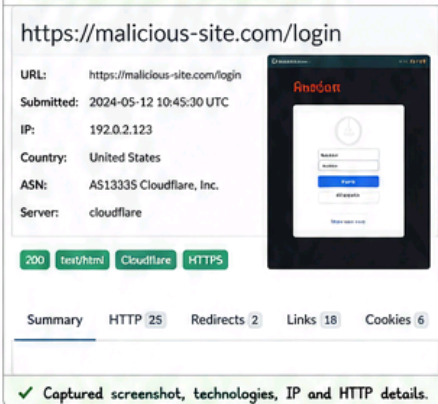
	VirusTotal	Scan URLs, IPs, files and domains.
	URLScan.io	Analyze website behavior, technologies and screenshots.
	AbuseIPDB	Check IP abuse reports and threat score.
	WHOIS	Get domain ownership and registration details.
	SecurityTrails	Search historical DNS, subdomains and relations.
	Cisco Talos	Get threat intelligence and reputation data.
	Shodan	Search for exposed services and open ports.
	crt.sh	Search SSL certificates and subdomains.

★ REAL EXAMPLES

1. VIRUSTOTAL - URL REPORT



2. URLSCAN.IO - ANALYSIS REPORT



3. WHOIS - DOMAIN INFORMATION

Domain Name: MALICIOUS-SITE.COM
Registry Domain ID: 2839483743_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2024-05-10T08:22:45Z
Creation Date: 2024-04-30T12:15:22Z
Registry Expiry Date: 2025-04-30T12:15:22Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Name Server: NS1.DNSPOD.NET
Name Server: NS2.DNSPOD.NET

★ IOC CHECKLIST

- ✓ URL
- ✓ Domain
- ✓ IP Address
- ✓ SSL Certificate
- ✓ WHOIS Info
- ✓ Hosting Provider
- ✓ DNS Records
- ✓ Subdomains
- ✓ Reputation Score
- ✓ Screenshots / Content
- ✓ Technologies Used



★ COMMON DNS RECORDS TO CHECK

RECORD	DESCRIPTION	EXAMPLE
A	Maps domain to IPv4 address	example.com → 192.0.2.1
AAAA	Maps domain to IPv6 address	example.com → 2001:db8::1
MX	Mail exchange records	example.com → mail.example.com
NS	Name server records	example.com → ns1.example.com
TXT	Text records (SPF, verification)	example.com → v=spf1 include:...
CNAME	Canonical name / alias	www.example.com → example.com

REPUTATION SCORE GUIDE (EXAMPLE)

SCORE RANGE	RISK LEVEL	ACTION
0 - 29	Low Risk	Monitor
30 - 69	Medium Risk	Investigate
70 - 100	High Risk	Block / Escalate



SOC INVESTIGATION TIPS

- ✓ Always verify the URL from multiple sources.
- ✓ Check domain age - newer domains are riskier.
- ✓ Look for suspicious redirects and hidden iframes.
- ✓ Analyze SSL certificate issuer and validity.
- ✓ Check website content and behavior using URLScan.
- ✓ Review historical data (Passive DNS, SecurityTrails).
- ✓ Correlate with threat intelligence feeds.
- ✓ Document all findings and IOCs for reporting.



EXAMPLE SUSPICIOUS INDICATORS

- ⚠ Newly registered domain
- ⚠ Hidden login / fake pages
- ⚠ IP with high abuse score
- ⚠ Hosting in bulletproof networks
- ⚠ Multiple subdomains pointing to same IP
- ⚠ Low reputation score on VirusTotal

“

Investigating a website is like peeling an onion - every layer reveals more truth.
Use the right tools, follow the workflow and trust the evidence.



13. CASE DOCUMENTATION & FORENSIC REPORTING



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

★ WHY DOCUMENTATION MATTERS?



- ✓ Provides a clear and accurate record of investigation activities.
- ✓ Supports findings and stands in court.
- ✓ Maintains chain of custody and evidence integrity.
- ✓ Helps in knowledge sharing and improving future response.
- ✓ Protects the analyst and organization.

★ INVESTIGATION NOTES – WHAT TO RECORD?

Incident Number	: INC-2024-0510
Analyst Name	: Saiprasanna M
Date & Time	: 10-May-2024 14:30
Incident Type	: Malware Infection
Systems Affected	: WIN10-WS-23
Evidence Collected	: RAM, Disk Image, Logs
IOCs Identified	: IP, Hash, Domain
Root Cause	: Malicious Email Attachment
Impact	: Data Exfiltration Attempt
Recommendation	: Patch, Monitor, User Training

★ CHAIN OF CUSTODY – SAMPLE FORM

CHAIN OF CUSTODY RECORD				
Case / Incident ID	: INC-2024-0510			
Evidence ID	: EV-001			
Description	: WD Hard Disk (1TB)			
Collected By	: Saiprasanna M			
Date & Time	: 10-May-2024 14:30			
Location	: WIN10-WS-23			
FROM (Name/Role)	TO (Name/Role)	Date & Time	Purpose	Signature
Saiprasanna M (Analyst)	Lab Evidence Room (Custodian)	10-May-24 14:45	Evidence Submission	
Lab Evidence Room (Custodian)	Senior Analyst (Reviewer)	10-May-24 16:20	Analysis	
Senior Analyst (Reviewer)	Case Lead (Approver)	11-May-24 09:10	Review	
Case Lead (Approver)	Legal Team	11-May-24 11:30	Court Submission	

Note: Maintain a continuous chain of custody for every evidence item.

★ SAMPLE FORENSIC REPORT LAYOUT



★ EVIDENCE DOCUMENTATION CHECKLIST

FIELD	DESCRIPTION / EXAMPLE
Evidence ID	EV-001
Description	WD Hard Disk (1TB)
Source System	WIN10-WS-23
Date Collected	10-May-2024
Time Collected	14:30
Collected By	Saiprasanna M
Tool Used	FTK Imager 4.7.1
Hash (MD5)	b6f22a1c8d4f5...
Hash (SHA256)	7e2c5f4a8d9b3c3f...
Storage Location	Evidence Locker #3
Purpose	Incident Investigation
Notes	Write Blocker Used

★ REPORT WRITING – BEST PRACTICES

- ✓ Use clear, simple and professional language.
- ✓ Stick to facts – do not include opinions.
- ✓ Cite tools, commands and data sources.
- ✓ Include timestamps and timezones.
- ✓ Attach screenshots, logs and timelines.
- ✓ Cross-reference artifacts and logs.
- ✓ Keep the report well-structured and concise.
- ✓ Proofread before final submission.

★ COMMON MISTAKES TO AVOID

- ✗ Incomplete or missing documentation.
- ✗ Breaking chain of custody.
- ✗ Not verifying hash values.
- ✗ Assuming instead of proving.
- ✗ Lack of timeline and correlation.
- ✗ Poorly organized or unstructured reports.
- ✗ Forgetting to update notes in real-time.

★ EVIDENCE TYPE EXAMPLES

TYPE	EXAMPLES
Volatile	RAM Dump, Running Processes, Network Connections, Logs
Non-Volatile	Disk Image, Registry Hives, Files, Databases
Digital	Emails, Chat Logs, Documents, Browser Data
Physical	Hard Drives, USB Devices, Mobile Phones
Cloud	Cloud Logs, S3 Buckets, Email (Google Workspace, O365)

★ CASE TIMELINE DOCUMENTATION (EXAMPLE)

TIME (UTC)	EVENT	DETAILS	SOURCE
10-May-24 14:15	Alert Received	Malware alert from EDR	EDR Console
10-May-24 14:20	Initial Triage	Confirmed alert on WIN10-WS-23	EDR / SIEM
10-May-24 14:30	Evidence Collection	RAM Dump collected using DumpIt	DumpIt Log
10-May-24 14:45	Disk Imaging	Disk image created using FTK Imager	FTK Log
10-May-24 15:20	Log Collection	Collected Event Logs, Sysmon Logs	Event Viewer
10-May-24 16:00	Analysis Started	Analyzing artifacts and timelines	Forensic Tools
10-May-24 18:40	Report Drafted	Initial findings documented	Investigation Notes

💡 SOC TIP

- ✓ Document as you investigate.
- ✓ Use consistent naming for cases and evidence.
- ✓ Store evidence securely and maintain access logs.
- ✓ Always preserve original evidence.
- ✓ A good report + good evidence = strong case.



“ A well-documented case is not just a report – it is the story of how the truth was found, step by step. ”